diarrheal stools by saline and iodine wet preparations (Figure 8.8).

Serodiagnosis: Immuno chromatographic strip tests and indirect immunofluroscence (IIF) tests are readily available. For antigen and antigen detection ELISA, Commercially available ELISA kits detects Giardia – Specific antigen.

Molecular methods: DNA probes and polymerase chain reaction (PCR) have been used to demonstrate parasitic genome in the stool specimen.

8.3.8 Treatment

Metronidazole and Tinidazole are the drugs of choice.

8.3.9 Prevention and Control

Giardiasis can be prevented and controlled by,

- Proper disposal of human faeces, maintenance of food and personal hygiene and health education.
- After using the bathroom and before eating, the hands should be washed thoroughly with soap and warm water. Boiling of water is the best and effective method in killing the viable cysts.
- To reduce the risk of venereal transmission, patients should avoid risky sexual behavior.
- No vaccine or effective chemo prophylactic drug is available for prevention of Giardiasis.

8.4 Tissue Flagellates – *Leishmania donovani*

The genus is named after the scientist Leishman, who first described the parasite in London in May 1903.

8.4.1 Geographical Distribution

Leishmania species is found in the Mediterranean, the Middle East, Africa and Asia including India.

8.4.2 Habitat

Leishmania donovani is an obligate intracellular parasite of human and other mammalian hosts. They are always found as intracellular amastigotes in the reticuloendothelial cells of the spleen, bone marrow, liver, intestinal mucosa and mesenteric lymph nodes of hosts.

8.4.3 Morphology

The parasite exists in two forms:

Amastigote: It is the form found in human and other mammalian hosts. They are found inside monocytes, polymorphonuclear leucocytes or endothelial cells. They are small, round to oval bodies measuring 2–3µm in length (Figure 8.8). They are also known as LD (Leishman donovan) bodies.

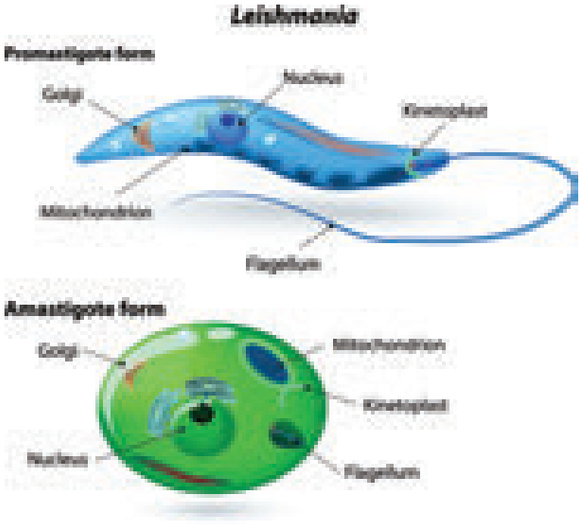


Figure 8.8: Promastigote and Amastigote form of Leishmania

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